

DIONIS LUDOVICK NASUWA

Full-Stack Software Engineer & Systems Developer

Location: Dar es Salaam, Tanzania (UTC+3) **Email:** nasuwadio36@gmail.com
Phone: +255 719 231 593 (WhatsApp) / +255 754 794 174 **GitHub:** github.com/dionis36
LinkedIn: linkedin.com/in/dionis-nasuwa

PROFESSIONAL SUMMARY

Computer Science student at the **University of Dar es Salaam (CoICT)** with practical experience in full-stack web development, relational and vector databases, systems programming, and enterprise IT infrastructure. Certified in **Full Stack Open (University of Helsinki, Grade 5/5)** and **Scrum Fundamentals Certified (SFC™)**. Experienced in building responsive web applications, RESTful APIs, and distributed monorepos using TypeScript, JavaScript, Python, Node.js, and React. Hands-on background in LAN setup, network switch configuration, and hardware diagnostics across 3 technical organizations.

EDUCATION & PROFESSIONAL CERTIFICATIONS

Bachelor of Science in Computer Science — College of Information & Communication Technology (CoICT), University of Dar es Salaam **Expected Oct 2026**
Core Coursework: Data Structures & Algorithms, Database Systems (SQL), Object-Oriented Analysis & Design (OOAD), Operating Systems & Linux Administration, Computer Networks, Systems Programming, Software Engineering.

Full Stack Open Certification July 2026 University of Helsinki • Grade 5/5 (Maximum Score) Comprehensive curriculum covering React, Redux, Node.js, Express, REST APIs, GraphQL, TypeScript, and web security.	Scrum Fundamentals Certified (SFC™) July 2026 SCRUMstudy • Professional Certification Validated knowledge in Agile principles, sprint planning, backlog management, and cross-functional team coordination.
--	---

TECHNICAL WORK EXPERIENCE

Software Engineering Collaborator — Mainstream Group (Technology Division) **March 2026 – May 2026** | Dar es Salaam
Technologies: TypeScript, JavaScript (ES6+), Node.js, RESTful APIs, Git, Agile Workflows

- Developed and integrated full-stack application features using **TypeScript** and **Node.js** across multi-module systems.
- Built and tested secure **RESTful API endpoints**, optimizing data exchange between client applications and backend databases.
- Participated actively in **Agile sprint workflows**, daily standups, and peer code reviews to ensure clean code standards and milestone delivery.

Software Development Trainee — UDICTI / StartHub Africa Partnership **July 2025 – September 2025** | Dar es Salaam
Technologies: Flutter, Dart, Django, PostgreSQL, OpenStreetMap API, REST APIs

- Collaborated on mobile app development and UI/UX layout design for the **Smart Cargo** digital logistics management system.
- Implemented dynamic rate calculation logic (distance, weight, fragility) paired with **OpenStreetMap** API for fare estimation.
- Integrated Swahili and English localization modules and assisted in database query optimization for real-time parcel status tracking.

IT Infrastructure & Network Optimization Trainee — TPC Limited **July 2024 – September 2024** | Kilimanjaro
Technologies: Managed Switches, Routers, Static IP Binding, LAN Troubleshooting, Hardware Maintenance

- Configured network switches, deployed wireless routers, and assigned static IP bindings to optimize enterprise LAN connectivity and stability.
- Diagnosed and resolved hardware and operating system issues across 50+ desktop workstations and network peripheral devices.
- Conducted systematic cabling, router diagnostics, and preventative maintenance routines to ensure high network uptime across departments.

KEY TECHNICAL & SYSTEMS PROJECTS

Plagode Platform — Digital Identity & QR Monorepo

TypeScript · Next.js · Fastify · Prisma · Docker

- Architected an 8-app distributed monorepo using **Turborepo**, coordinating shared UI component packages and backend services.
- Implemented a stateless authentication layer utilizing **RS256 JWT tokens** for asymmetric cryptographic verification across subdomains.
- Configured containerized environments with **Docker** and managed database migrations with **Prisma ORM** on PostgreSQL.

INTLEQ — Local LLM RAG & Semantic Vector Search System

Python · FastAPI · Ollama · pgvector · PostgreSQL

- Developed an offline-first **Retrieval-Augmented Generation (RAG)** application executing local LLM inference via **Ollama (llama3.2)**.
- Built asynchronous REST API endpoints with **FastAPI** and performed cosine similarity vector queries using **pgvector** in PostgreSQL.
- Implemented document chunking, automated schema migrations with **SQLAlchemy**, and containerized the vector database with Docker.

PataSpace — Real Estate Marketplace & Property Platform

Django · React · Flutter · PostgreSQL · Leaflet Maps

- Built a full-stack rental platform with **Django 5**, **React**, and **Flutter** featuring mobile money integration concepts.
- Integrated **Leaflet Maps** marker clustering and field-agent GPS photo verification workflows for authenticated property listings.
- Implemented notification pipelines utilizing SMS APIs (Africa's Talking) and PostgreSQL indexing for rapid location-based searches.

Strata (Final Year Project) — AST Code Analysis & Visualization Engine

Python · AST Parsing · Graph Theory · Streamlit · Docker

- Engineered an analytical code engine parsing source codebases into **Abstract Syntax Trees (ASTs)** and **Directed Acyclic Graphs (DAGs)**.
- Applied **Strongly Connected Components (SCC)** clustering algorithms to identify circular dependencies and legacy architectural bottlenecks.
- Constructed an interactive **Streamlit** dashboard visualizing structural class dependencies and system complexity metrics.

TECHNICAL SKILLS & CORE COMPETENCIES

Programming Languages:	JavaScript (ES6+), TypeScript, Python, SQL, Dart, HTML5, CSS3
Frontend & Mobile:	React, Next.js (App Router), Flutter, Redux Toolkit, Tailwind CSS, Responsive Web Design, Streamlit, Vite
Backend & APIs:	Node.js, Express.js, Fastify, FastAPI, Django, RESTful API Development, GraphQL, WebSockets, JWT Authentication
Databases & ORMs:	PostgreSQL, MySQL, Redis, Supabase, pgvector, Prisma ORM, SQLAlchemy, Relational Schema Design
DevOps, Tools & Cloud:	Git & GitHub, Docker, Turborepo (Monorepos), CI/CD Pipelines, Linux (Ubuntu, Fedora), Bash, Nginx, Fly.io, Vercel, Postman
IT Infrastructure:	Local Area Networks (LAN), Switch & Router Configuration, Static IP Binding, Network Troubleshooting, Hardware Diagnostics
Methodologies:	Agile / Scrum (SFC™ Certified), Object-Oriented Programming (OOP), Test-Driven Development (TDD), AST & Graph Analysis

SPOKEN LANGUAGES

English: Fluent / Professional Working Proficiency **Swahili:** Native Proficiency